INCH-POUND
MIL-DTL-12109/21B
11 January 2011
SUPERSEDING
MIL-DTL-12109/21A
20 August 1998

DETAIL SPECIFICATION SHEET

SOLENOID, ELECTRICAL; CLASS 2 (MACHINE GUN) 7.62 MM, 24 VOLT DC, 8 AMP MAX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-12109.

APPLICABLE DOCUMENTS:

Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-12109 - Solenoid, Electrical; Gun-Firing Mechanisms, General Specification For

(Copies of these documents are available from https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DRAWINGS

7792103	-	Gasket
7792104	-	Cable Assembly
75020922	-	Connector Assembly

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(Copies of these drawings are available from DAMI_STANDARDIZATION@conus.army.mil or U.S. Army RDECOM, Tank Automotive Research, Development and Engineering Center, ATTN: RDTA-EN/STND/TRANS MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM20995	-	Wire, Safety or Lock
NASM24677	-	Screw, Cap, Socket Head, Hexagon, Drilled Alloy Steel (UNC-3A)
NASM33540	-	Safety Wiring and Cotter Pinning, General Practices For

(Copies of these documents are available from http://www.aia-aerospace.org or from Aerospace Industry Association, 1250 Eye Street, N.W., Suite 1200, Washington, D.C. 20005-3924.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI J STD 004,005,006	-	Flux, Soldering, Liquid, Paste Flux, Solder Paste, and Solder-Paste Flux; General Specification For

(Copies of these documents are available from www.ansi.org or ANSI Customer Service Department, 25 W. 43rd Street, 4th Floor, New York, NY 10036.)

ASTM INTERNATIONAL

ASTM D412	-	Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers – Tension
ASTM D429	-	Standard Test Methods for Rubber Property Adhesion to Rigid Substrates
ASTM D430	-	Standard Test Methods for Rubber Deterioration – Dynamic Fatigue
ASTM D2000	-	Standard Classification System for Rubber Products in Automotive Applications

(Copies of these documents are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, PO Box C700, West Conshohocken, PA, 19428-2959)

REQUIREMENTS:

Requirements for submission of first article shall be as specified in the contract.

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Stroke shall be 0.165 to 0.247 inches (in.) (0.42 to 0.63 centimeters (cm)).

Weight shall not exceed 2.25 pounds (lb) (1 kilogram (kg)).

Solenoid shall functionally operate against the static forces specified in figure 1, at 20 and 30 volts direct current (Vdc).

The current shall not exceed 8 amperes maximum.

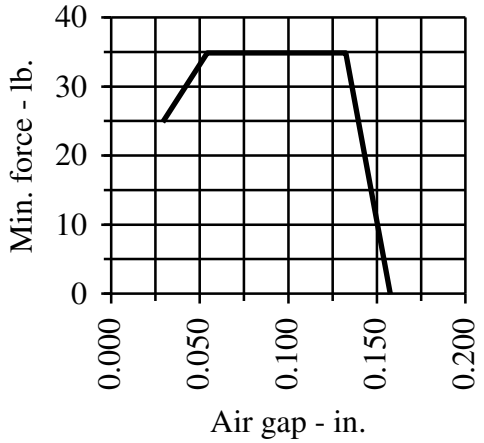


FIGURE 1. Static force - air gap curve.

Solenoid shall meet the static force air gap curve shown in figure 1.

Solenoid shall be capable of firing 10 000 rounds.

Solenoid shall withstand a dielectric withstanding voltage potential of 1000 volts root mean square. A leakage current of 100 micro-amperes max is permitted.

VERIFICATION:

The first article shall be submitted for inspection in accordance with contract requirements. The first article shall be tested in accordance with the tests specified herein and the testing provisions of MIL-DTL-12109.

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a. The stroke test shall be performed during functional (operating) test.
b. Weight shall be confirmed by any approved method.
c. Testing for dielectric withstanding voltage shall as follows:

Test voltage of 1000 volts root mean square at 60 Hertz (Hz) in frequency and shall approximate, as closely as possible, a true sine wave in form. The voltage shall be applied between coil terminal and ground metal parts for at least 60 seconds. Test voltage shall be raised from zero to 60 Hz in frequency at a rate of approximately 500 volts per second.

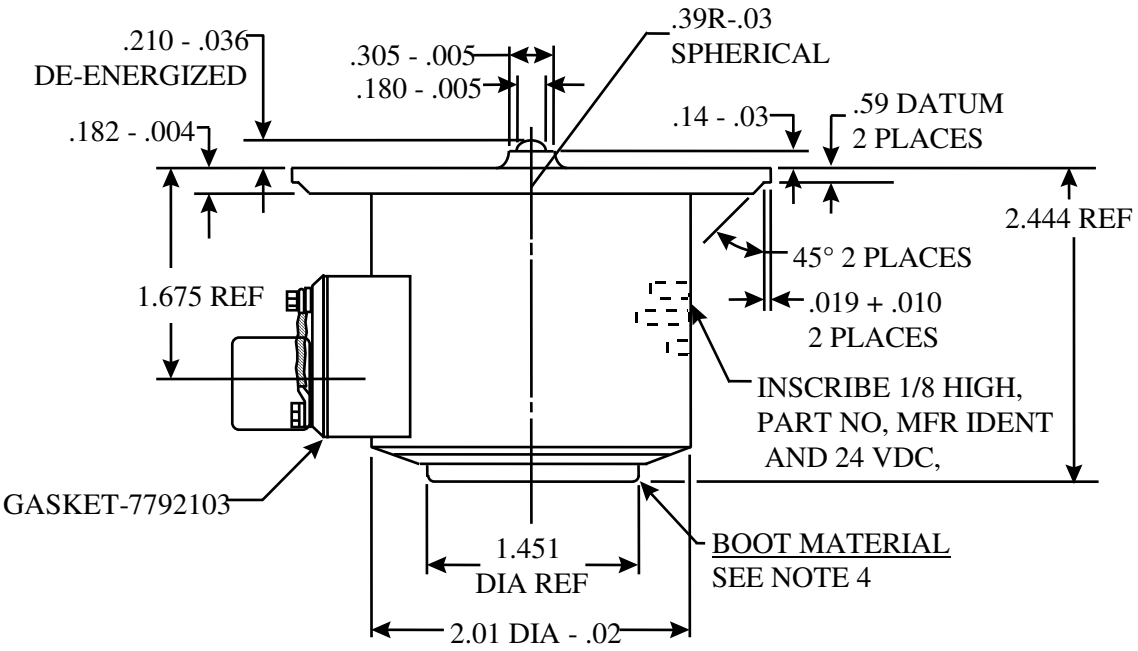
NOTES:

PART OR IDENTIFYING NUMBERS (PIN):

M12109/21-1 (Former part number 7791685 with weapon designation M60C)
M12109/21-2 (Former part number 75F20924 with weapon designation M60E2)

CHANGES FROM PREVIOUS ISSUE: The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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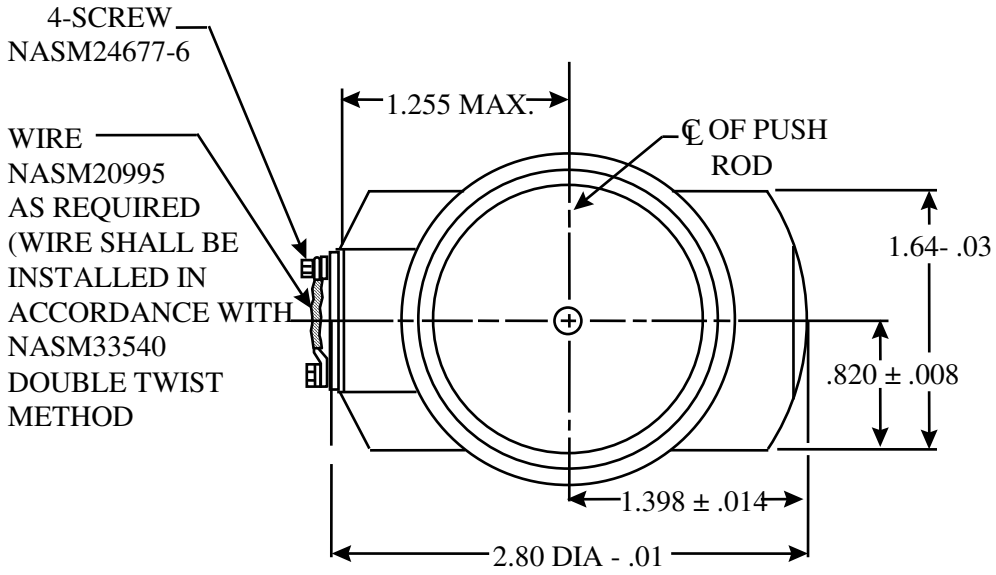
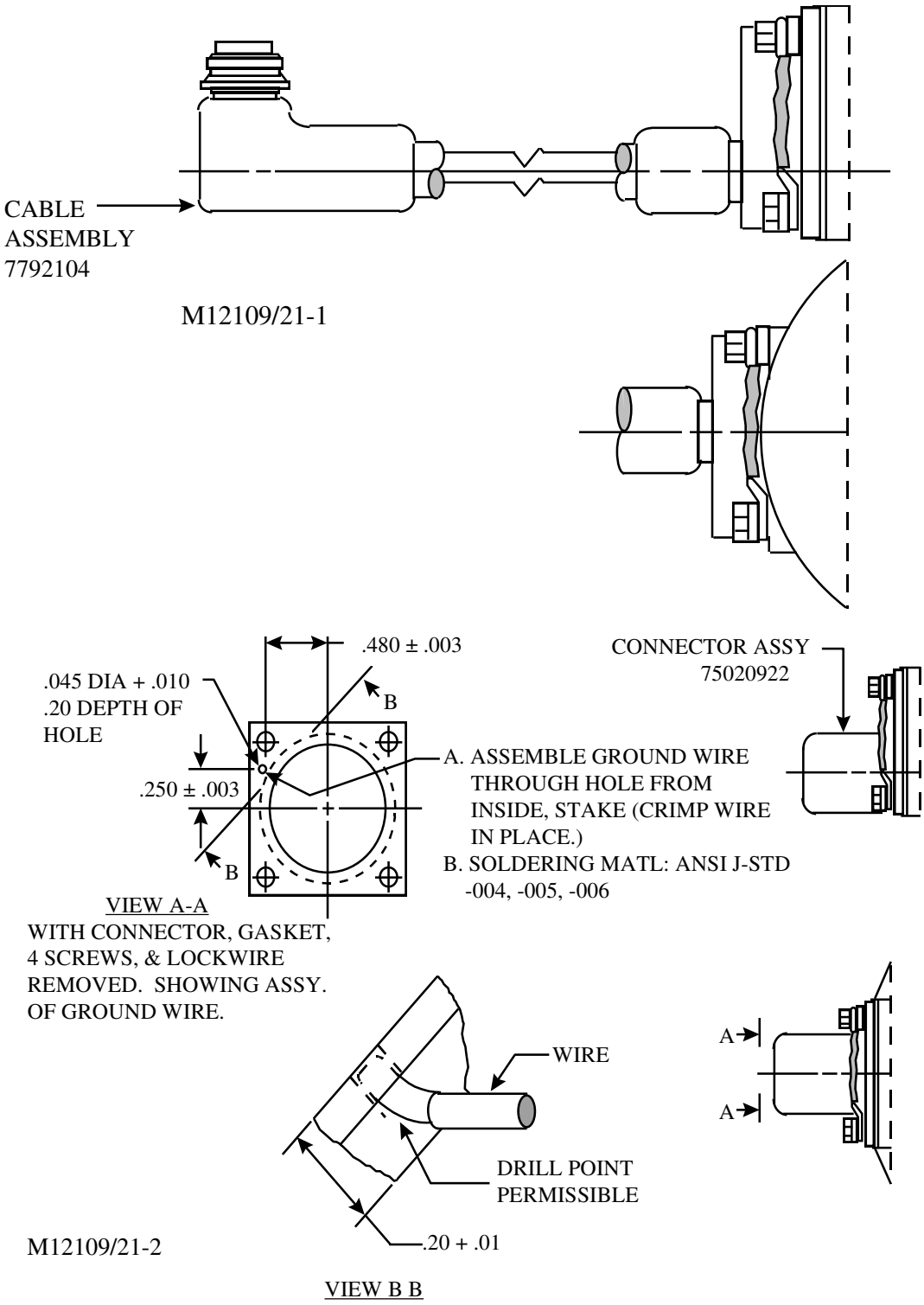


FIGURE 2. Solenoid characteristics.

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FIGRE 2. Solenoid characteristics - Continued.

NOTES:

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1. This drawing is for interface and envelope requirements and is not intended to limit design.
2. Dimensions are in inches. Unless otherwise specified, tolerances are ±.010 on decimals, ±2° on angular dimensions.
3. REF = Reference
4. The solenoid boot shall be made per the following properties:
   ASTM D2000 M2BC 614 B13 C32 F17 Z1 Z2 Z3
   Z1 = H2 (38) IAW ASTM D430 METHOD B
   Z2 = 350% MAX. ULTIMATE ELONGATION IAW ASTM D412 DIE C
   Z3 = K24, 3.5 KN/M WIDTH IAW ASTM D429 METHOD B, 100 °C

FIGURE 2. Solenoid characteristics - Continued.

Custodians:
Army - AT
Navy - OS
Air Force - 85

Review Activities:
Army - AR, EA
Navy - MC
Air Force - 99
DLA - CC

Preparing Activity:
Army - AT

(Project 5945-2011-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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